

ALMA MATER STUDIORUM – UNIVERSITÀ DI BOLOGNA

DEPARTMENT OF STATISTICAL SCIENCES

“PAOLO FORTUNATI”

First Cycle Degree / Bachelor in Statistical Sciences

Curriculum Stats&Maths

The Effect Of Weather Shocks On U.S. Industrial Production:

A Local Projection Approach

Econometrics

Presented by:

Nikola Stevanović

0001068141

Supervisor:

Prof. Emanuele Bacchiocchi

SESSION I, July 2025

ACADEMIC YEAR 2024 /2025

1 Abstract

This study examines the short to medium-term impact of weather-related disasters on the economy in the United States by applying statistical techniques to a comprehensive dataset spanning three decades. Unlike other studies, which more often focused on different types of localized shocks, this study will address the problem in the dataset, which measures the effect over all fifty states. The data used in the research has been collected by the National Centers for Environmental Information over 30 years, starting in 1990, as well as the Federal Reserve Bank of St. Louis. The estimation of the effects is going to be the product of econometric analysis, where the estimates will be deduced from impulse response functions, which in turn are going to be estimated using local projections. This research will further prove that weather disasters exhibit a non-linear effect on the economy. The effect has a non-linear pattern as seen through our models. Furthermore, we can conclude that the significance of the effect deeply depends on the length of the predictions and the values it has been predicted.

Implications of this study could be in a variety of fields, predominantly in policy analysis as well as environmental and economic studies. Furthermore, this research could be extended in aspects such as trying to estimate the effect on non-aggregate data, as well as seeing whether disaster impacts different economic sectors.

Contents

1	Abstract	1
2	Introduction	3
3	Literature Review	4
3.1	Economic Theories on Market Dynamics and Weather Shocks Effect	4
3.2	Empirical Methodology	6
4	Data and Methodology	8
5	Preliminary Times Series Analysis	11
6	Results and Discussion	13
6.1	Lag Selection	13
6.2	Linear Local Projection	13
6.3	Non-Linear Local Projection	15
7	Conclusion	17
8	References	18

2 Introduction

In today's day and age, there is ever-growing interest in sustainability, which arises mainly due to the constant thinning of natural resources, and above all, due to the fear of climate change. Prevention of possible extreme weather shocks and possible impact on the market dynamics is just another of the benefits reaped from investing in sustainability. In the following thesis, we are going to explore the relationship between climate change and financial impacts. We will explore how the two are intertwined, using econometrics models such as local projections (LP) as a means of estimating impulse responses through impulse response functions (IRFs). We will do so on a scale of the whole United States of America, by taking into account many major weather shocks over the past decades. The data has been collected by the National Center for Environmental Information and encompasses the major weather disasters spanning from 1990 to 2019, as well as the Federal Reserve Economic Data database managed by the Federal Reserve Bank of St. Louis. The cut-off in 2019 was done so that the research would not be affected by the possible outliers during the pandemic of COVID-19. The collection of data on weather disaster-induced damages has been mostly conservative, as it is explained in the data chapter. The disasters in question are drought, flooding, freeze, severe storms, tropical cyclones, wildfires, and winter storms. The financial impacts caused by the weather disasters represent the additional expenses that occurred only because the disasters happened. NCEI also provides data on casualties of each weather disaster. Also, we are going to make use of the monthly Industrial Production Index obtained from FRED

As we previously explained, the dataset and the process that was used for the collection of the data should also provide the tools with which we will explore the said data. Our econometric models are going to be implemented using R software and the `lpirfs` package created by Phillip Adämmer.

This data and tools, together with the idea that there is a relationship between weather disasters and the Industrial Production Index, will serve as the backbone of our work. This stems from the idea that the possible logistical difficulties during weather-intense times, possible unavailability of human capital, weather damage to production centers, and a shift of government interest affect industrial production. Other channels of impact will be studied more deeply in the literature chapter

Such a topic might be of great interest not only to econometricians but also to policymakers, as it may show the impact climate change has, through weather shocks as its proxy, on economic loss. It could pose as a representation of urgency when discussing climate change. It could further signify the impact of extreme weather amplitudes on the US economy and, by extension, on a global scale. This conclusion could be made due to multiple factors. The US is the sole global power in today's economy; any more impactful hits on its economy could be strongly felt in other economies worldwide. It should not diminish its environmental and geographical diversity, which could show the effect of previously named disasters on different environments. This could

be an example for the policymakers, econometricians, and businessmen from countries with similar climates and environments, on how to cope with these disasters, and what are expected effects on the economy.

The thesis is split into six parts: Introduction, where we will shortly go through the main problems behind the thesis, the data used for the thesis, its justification, and the structure of the thesis. Literature review, which will include an explanation of market dynamics due to the weather effects, and empirical procedures needed for the research. In this section, we are also going to mention previous research and analyze it more in-depth. We should also discuss possible different methodologies used for local projections of the impulse response function. The third part is going to be the representation and description of data. Including its collection, organization, and presentation. The fourth part will include an empirical analysis of the data, which will include preliminary statistical analysis and different implementations of local projections to obtain estimates of impulse response functions. The final chapters are results and discussion about the previously obtained outcomes from the statistical analysis, and the final part is going to be the conclusion of the thesis, where we will summarize the findings, their impact, the limitations, and possible future research.

To my knowledge, there have not been previously done studies using this methodology to estimate the effects of weather disasters on the economy. Hence, this study should cover the technical basics when it comes to utilizing the local projection for this cause and could be used as a groundwork for future studies.

3 Literature Review

3.1 Economic Theories on Market Dynamics and Weather Shocks Effect

It is a known fact that every year we experience an increase in global temperature. Almost every summer month, the mainstream media will inform us of the hottest month measured. According to the UN, 2024 has been the hottest year on record, reaching 1.5 degrees above pre-industrial temperatures. The collateral damage of global warming includes natural catastrophes such as hurricanes, droughts, and wildfires, and increased tropical storms. Which shall be denoted further in this article as weather disasters or shock variables. Some are a product of the heated surface of the global sea, which in turn leads to stronger sea-bound winds, and others as a consequence of increased hot vapor in the atmosphere, which is used as a fuel for tropical storms by the USGS. As stated by Eickmeier, Quast, and Schöler (2024), extreme weather shocks should be treated as ambiguous because models based on the neoclassical approach of economics predict weather shocks alongside any other type of nuance to have only an instantaneous effect on both the capital and labor markets. The neoclassical economic approach is one in which the utmost effort is given to the allocation of resources at a given time point as argued by Lorek and Spangenberg(2005). So, in turn, we could say that it is a broad theory that underlines today's

economy, and it is based on the fact that the driving force of economic processes is supply and demand.

According to the neoclassical approach, short-term effects should not have long-term effects on the economy and its growth. Following the research of Eduardo Cavallo, Sebastian Galiani, Ilan Noy, and Juan Pantano (2013), the large-scale disaster effect on long-term growth significantly happens only in occurrences where the events are followed by radical political revolutions, and when controlled for those, the effect diminished. Although this finding might seem to limit the expected persistence of the disaster effect, we should note that economic theory does not fully answer this question. These results, although strong, are still empirical and are model-dependent, making them prone to change, based on the model used as well as the economic parameter studied. Therefore may yield different outcomes when estimating the effects of disasters. Furthermore, the data on which the models are computed differs. Instead of observing an effect on GDP, we mainly focus on the Industrial Production Index (IPI) as a measure of economic activity. This allows us to measure the effect in a shorter period, compared to the long-term effect studied in the previously named paper.

Nonetheless, in our approach, the response is going to be modeled on the weather disasters as exogenous variables. Meaning the variables in question should be unaffected by other foreign influences. It should also be stated that one of the key factors when estimating the potential impact of weather shocks is the region of impact. As Eickmeier (2024) suggests, the bigger the region of impact, the chances that natural disasters cause changes to aggregate prices. It is also possible to deduce that the increase in size of the region could raise the spillover effect. Which could further affect other states, and possibly later aggregate IPI. However, in our study, we are not going to account for the spillover effect, as we are working with aggregated data, and the nation-based spillover effect does not play an important role.

A further property of extreme weather shocks and climate change is that they could affect the market dynamics through different channels of impact. For instance, the most common manner is the destruction of capital, as in facilities, farmland, and further infrastructure. Prolonged weather conditions such as intense droughts and winter storms could harden the act of labor, affecting countries under development more drastically. It would be hard to test for these assumptions, not only because of the development status of the US but also because of finding appropriate metrics that show how labor could be affected. However, the effect on labor will be seen in our study through the total financial losses and IPI. As argued by Acevedo, Mrkaic, Pogacheva, and Topolalova(2020) prolonged increased temperatures and precipitation both affect production quadratically, meaning where the temperature is low, and there is no precipitation the increase in the same leads to prosperity, whereas if there is already enough of two, further may cause slowing down of production, and negatively affect market dynamics. This statement highlights the importance of observing market dynamics before estimating impulse responses at a certain time frame. Namely, the weather conditions and temperature before extreme weather shocks in certain regions might influence the significance of the weather shock itself. Another important

channel of impact is human capital. A common occurrence is that, due to the extreme conditions, people are forced to flee their homes and workplaces. Ferreira(2024) indicates that it may cause mental health problems that will, in the future, affect the efficiency of workers. However primary risk is the high morbidity and mortality that may occur immediately after the shock. All of this may cause the displacement of human capital and further alterations in tracked dynamics.

For this thesis, it is important to observe how weather disasters affect, among others, prices, consumption, and government spending to better understand the dynamics of the market. Research by Ferreira (2024) showed that both government and private capital are affected by natural disasters. A common occurrence is that government debt increases, due to the public spending from the disaster relief funds. Furthermore, the prices of energy rose shortly after the disasters, alongside the prices of food, both according to the consumption index. This is a product of the increased costs in transportation, predominantly, followed by production, as written by Eickmeir(2024). This again shows the impact of weather shocks through the previously explained channel of impact. Our goal is going to be to further analyze the relationship between weather shocks and IPI, predominantly showing the effect of the manufacturing, mining, and utility sectors, to further understand their causal relationship.

3.2 Empirical Methodology

In recent years, there has been a surge in the use of local projections, as opposed to vector autoregressions, as an econometric approach. The beginning of this trend could be attributed to the work of Jordà (2005), on which this chapter will be closely based.

To appropriately use an econometric technique such as impulse response analysis using local projections requires an understanding of fundamental concepts and their econometric applications.

First of all, it is of utmost importance to understand what impulse response functions and local projections are. According to Møller and Wolf (2021), the impulse response functions, or IRF, are dynamic responses to a structural shock. In a sense, it represents the effect of exogenous variables, which in turn represent external shocks, which cannot be affected, on certain economic parameters of interest. Secondly, Local projections or LPs represent an alternative method for estimating the IRFs and are often used instead of vector autoregression or VARs. According to Jordà (2005), the idea on which LPs are based is that we run the sequence of regression, on a certain shock, at each period of interest, rather than increasingly larger horizons; it is done with vector autoregressions (VARs).

There exists a set of properties that gives good reasoning for the ever-emerging interest in LPs. Jordà (2005) highlights that the LPs are, in general, easier to compute, given that they can be estimated using simple least squares. He further highlights that inference could be attained without the use of asymptotic properties, which will eliminate the problem of missing data. Finally, they are flexible models, as the estimates are not usually affected by non-linear

misspecification.

After stating the basic definition and certain advantages of LPs, we should further focus on a methodological framework upon which this model is conducted.

In the view of Jordà (2005), there exist two most widespread methods for the computation of Impulse responses. The first one is a two-step procedure that makes use of the Wold decomposition. However, Jordà argues, that complications may arise using this approach due to the data-generating process. Jordà suggests that the better way to obtain an impulse response estimate is by taking the difference between two estimated predictions.

$$IR(t, s, d) = \mathbb{E} [y_{t+s} \mid v_t = d; X_t] - \mathbb{E} [y_{t+s} \mid v_t = 0; X_t], \quad s = 0, 1, 2, \dots \quad (1)$$

The formulation (1) represents the difference between two predictors, which differ in the vector of regressors v_t , where one is a zero vector, and the other contains relevant shocks. This approach is suitable only when the said model correctly represents the data-generating process. Furthermore, Jordà (2005) suggests that in the linear case, IRF can be simply presented as:

$$IR(t, s, d_i) = B^s d_i, \quad s = 0, 1, 2, \dots \quad (2)$$

In this formulation B^s is the matrix of coefficient estimates at each horizon s , and d_i is the vector of relevant shocks. Here lies the strength of this approach. Where, instead of recursive estimation in vector autoregression, each forecast horizon has its regression. We should also note that in this model, the regressors are based on information before time t . While the errors represent a further deviation in our model. This makes the regressors and the errors uncorrelated.

Aside from the benefits of local projections, like any other statistical model, they face certain challenges. For instance, consistency and efficiency, which Jordà (2005) proved through Monte Carlo methods. Namely, the consistency is not significantly affected by the change of variables, lags, and the size of the horizon. For which the models do not need to be estimated jointly. Another difficulty that's often faced, considering the local projection models, is a trade-off between bias and variance. This could also be interpreted as a difference between VAR and LP at different sizes of horizons. Namely, according to Møller and Wolf (2024), when estimating a horizon bigger than the lags of the variables, VAR extrapolates and could introduce bias in the case of a more complex data structure. LPs, on the other hand, estimate impulse responses at each horizon. Hence, because it uses all autocovariance, it is more flexible than two. However, the variance of estimates given by LPs is bigger than that of VAR. However, one should keep in mind that according to recent discoveries from Plagborg-Møller & Wolf (2021), these two methods prove to perform similarly when the lag goes to infinity. So, one should choose between the two only based on lag length, whose maximum can be computed by information criteria. Hence, the size of the lag should be the product of the minimization of either AIC or BIC.

After understanding the fundamental concepts and how they operate, we will need to describe the practical setting in which the model will be implemented. This shall consider the require-

ments regarding data and its corresponding software implementation. In our case, where we use (1) as a definition for our impulse responses, it is defined without reference to the unknown DGP, meaning that a fixed DGP is not assumed. As stated by Li, Møller, and Wolf (2024), according to the DGP, we can devise three ways of obtaining impulse response estimates.

- Observed shock identification. In which the impulse response is a product of observed endogenous variables and shocks. The biggest issue here is the measurement error.
- IV/proxy identification. Where the individual does not observe shock directly and instantaneously, but rather through a proxy, with a certain random error.
- Recursive identification. The results of this approach are similar to those of Observed shock identification and differ solely due to random error and robustness.

The LP will be implemented in R using the `lpirfs` package, whose details will be discussed more thoroughly during the empirical analysis of the implemented model

4 Data and Methodology

In this chapter, the discussion is going to be centered on the data used for our research as well as the methods we applied to adjust and transform it so it is suitable for the models at hand.

First of all, we should mention the origin of the data and the methodology by which it was collected. The data have been collected for over 40 years, starting in 1980, by the National Centers for Environmental Information (NCEI). Over this period, there had been 403 extreme weather disasters, cumulating nearly 3 trillion USD in damages, and causing nearly 17 hundred deaths across the United States. According to NCEI, disasters it is considered drought, flooding, freeze, severe storms, tropical cyclones, wildfire, and winter storms. As stated by NCEI, their goal is to track climate events in the US over time and present their effect on the economy. Over the past years, they worked on methodology, and together with the team of scientists, they concluded that, in general, they are underestimating the effects of weather shocks. Their reasoning for the impact on the economy is mostly based on the developments in the vulnerable areas, as well as an increase in population and material wealth. In the end, it is also important to understand that, as a definition of financial loss, NCEI considered the amount of dollars that would have been accumulated had the event not taken place. These estimates are somewhat conservative compared to true losses. Furthermore, this data is available corresponding to each of the disasters in the timeframe in which NCEI is conducting research. We aggregated on a nationwide monthly frequency, resulting in a total of 360 observations over the period studied.

However, it should also be noted that due to the nature of data collection, NCEI is prone to miss weather events, whose impact is not equally devastating economically. Which might lead to misspecification, especially in underdeveloped and poor areas.

On the other hand, the monthly data on the Industrial Production Index or IPI, as previously stated, has been collected from FRED. Which is a database organized and maintained by the Federal Reserve Bank of St. Louis. It has been operational since early 1990, according to FRED. Aside from IPI, they cover many different economic and social parameters, usually collected by Government agencies.

We should now present the variables we are going to use in our model and which role each of them is going to portray. As our dependent variable in our model, we are going to utilize the Industrial Production Index obtained from Fred. It will help us track the effect of weather disasters on the economy. This index provides a measure of real economic activity, especially in the manufacturing, mining, and utilities sectors. These sectors are prone to being affected by natural disasters, making them the perfect dependent variable of our model. Furthermore, this type of data has similarly previously been used in the study by Felbermayr & Groschl, to show more short-term economic activity.

This IPI data has a monthly frequency, pairing well with our monthly aggregated disaster data, and will span from January 1990 to December 2019. This data will help us broadly summarize the effect of natural disasters in an empirical way that has not been seen in many implementations.

In our models, we make use of a switching variable. Similarly to the approach of Auerbach and Gorodnichenko (2013), we use regime-switching models. To estimate the non-linear effect disasters have on the economy. The switching variable z_t is defined using the Hodrick-Prescott filter with parameter λ equal to 10000, similar to the previously named paper. Hodrick-Prescott filter is used to obtain a smoother representation of time series data according to Hodrick, Robert; Prescott, Edward C. (1997). This is done so, our non-linear model would not be affected by possible unusual observations. Furthermore, the impulse response functions will vary depending on whether the economy is in expansion or recession. As a switching variable and a measure of regime, we are going to consider the unemployment rate. The regime weight is determined by the logistic function defined as:

$$F(z_{t-1}) = 1/(1 + \exp(-\gamma z_{t-1})) \quad (3)$$

Where z_t switching variable and γ is the smoothing parameter. Which we consider to be equal to 15. The model will then be redefined as:

$$Y_{t+s} = (B_0)^s + (B_1)^s(1 - F(z_{t-1}))d_1 + B_1^s F(z_{t-1})d_2 \quad s = 0, 1, 2, \dots \quad (4)$$

The switching variables are lagged to ensure endogeneity. Namely, we want them to be independent of the disasters, and hence, we base our regime on the last switching variable before the disaster. The interpretation is that if our regime weight is approximately 1, we have a recession, and if it is approximately 0, we have expansion.

Alternatively, we implemented a threshold based on Sahm (2019). She suggests that if the

difference between a 3-month average and a 12-month minimum for this average exceeds 0.5, we have a recession. If this type of threshold is applied as a switching variable without using an HP filter and logistic function, we achieve interesting results however, their validity comes into question, as there have not been many studies assuming this method.

To obtain one of our regressors, we need to subject our data to some manipulation, which was closely based on the work of Bacchiocchi, Bastianin, and Moramarco (2024). Their work introduces the use of state-level dummy variables as weather proxies, which were defined as:

$$emergency = \begin{cases} 1 & \text{if } emergency \text{ happened} \\ 0 & \text{if } otherwise \end{cases}$$

To account for the intensity and spread of the disaster, we used a variable called "Disaster". This variable represents the count of disasters in each state. It could be understood as a nation-based aggregate of the emergency dummy. It is equal to one if any of the following disasters occurred in a given month across the U.S.: drought, freeze, flooding, severe storms, wildfires, tropical cyclones, and winter storms. Further interpretation could be as a broader measure of disaster frequency.

To better estimate the magnitude of the effect of weather disasters, we also use financial losses and casualties. Both of them have been collected from NECI, and we previously defined financial losses.¹ The casualties variable is the number of people fatally wounded as a consequence of weather disasters. All of these variables will be used in the model to estimate their effect on IPI in a one-year time frame.

¹The definition of financial loss, NCEI considered the number of dollars that would have been accumulated had the event not taken place

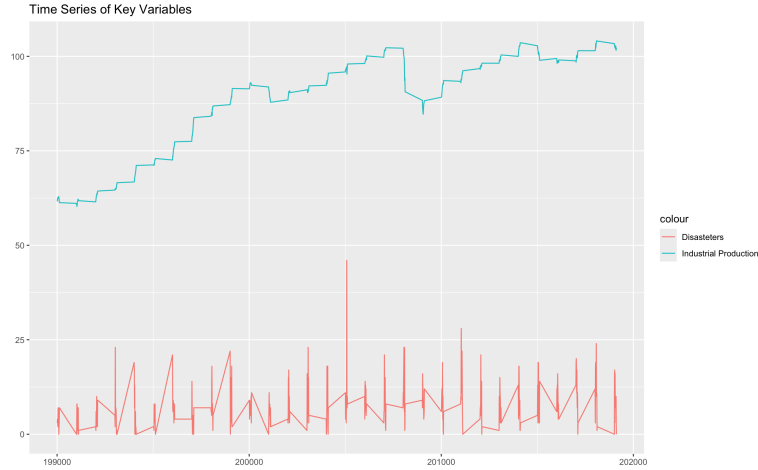


Figure 1: Time Series Plot of IPI

5 Preliminary Times Series Analysis

As a part of preliminary time series analysis, we take a deeper look into descriptive statistics, time series plots, stationarity test, ACF and PACF, variance and volatility, and correlation between series. We begin by reporting summary statistics for each of the variables in our dataset. The range of variables is different from one variable to another. Each of the variables shows a substantial standard deviation, indicating high data variability. The high value of the mean, followed by a big standard deviation, and their difference from the maximum suggest that the variables are right-skewed.

Table 1:

Statistic	Mean	St. Dev.	Min	Median	Max
Casualties	172.606	820.577	0	26	13,038
Financial_losses	34,649.980	117,127.300	0.000	7,180.600	1,446,509.000
Disasters	7.600	5.919	0	7	46
Affected_Counties	118.960	199.778	0.000	58.000	2,862.000
NA	1.863	3.045	0.000	1.133	44.781
INDPRO	88.395	13.280	60.297	92.336	104.104

These results suggest non-linearity in our variables and potential Heteroskedasticity, which will justify the use of the models in the later part of the thesis.

In the time series plot of disasters (Figure 1) and industrial production, we see an increasing trend over time for industrial production, while disasters do not have any kind of obvious trend. In both, we can spot seasonal behavior. In the disaster plot, this might be the hurricane season or winter months when disasters are more likely. On the other hand, we have industrial production whose cycles might be due to the economic cycles. We also see periods of more erratic movements, such as the economic crisis of 2008 and Hurricane Katrina in 2006.

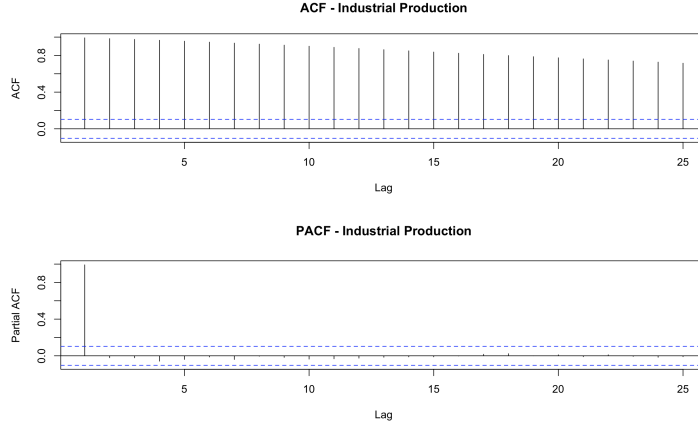


Figure 2: Autocorrelation function (ACF) and Partial Autocorrelation function (PACF)

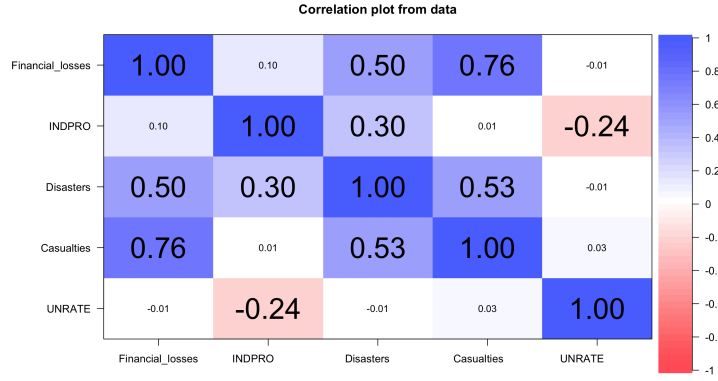


Figure 3: Correlation plot of variables

The results of autocorrelation (ACF) and partial autocorrelation (PACF) plots suggest that Industrial Production has an AR(1) structure.

After conducting the Augmented Dickey-Fuller (ADF) test, we see that the null hypothesis is not rejected, suggesting that the series is non-stationary.²

statistic	parameter	alternative	p.value	method	data
-2.00	7.00	stationary	0.58	Augmented Dickey-Fuller Test	IPI

Table 2: Augmented Dickey-Fuller test

The correlation plot further testifies to the independence of our switching variable, as it is not correlated with other regressors. We further see a moderate relationship between disasters and industrial production, however, these preliminary results should not be taken seriously as correlation ignores the dynamic structure of data and should be mistaken for a causal relationship

²This is expected, as previously commented on the trends of the data. Because of this, we are advised to take the first difference of our IPI dataset. This could eliminate non-stationarity, which will in turn allow us to attain better model predictions.

between the two variables. We can further spot relation positive relations between other variables. However, considering that we are not using a multiple-variables model, endogeneity should not be a problem.

6 Results and Discussion

In this chapter, we implemented the local projections to obtain impulse response functions. This will let us analyze the dynamic effect of weather disaster shocks over time.

6.1 Lag Selection

We first devised algorithms in which we will run the models. In the loop going from one to twelve lags, we are going to collect the values of the sum of squared residuals for each estimation at each lag. This will allow us to compute the BIC for each of the lags. We then considered the lag with the smallest BIC to be the optimal selection for the lag of our models. We chose the size of loops to be 12, as we believe that is most suitable for this type of monthly data.

6.2 Linear Local Projection

Naturally, we first had to create a new data frame that would store endogenous variables. Then, following the work of Jorda, we decided to implement linear local projections. As shown by the previous results of the algorithm, we choose lag 1, meaning that we look for the immediate effect of weather costs and deaths. In our function, we further set the trend to 0, as our model is linear and does not assume non-linear trends. We try to estimate the effect of disasters, financial losses, and casualties for the future 12 horizons on Industrial Production.

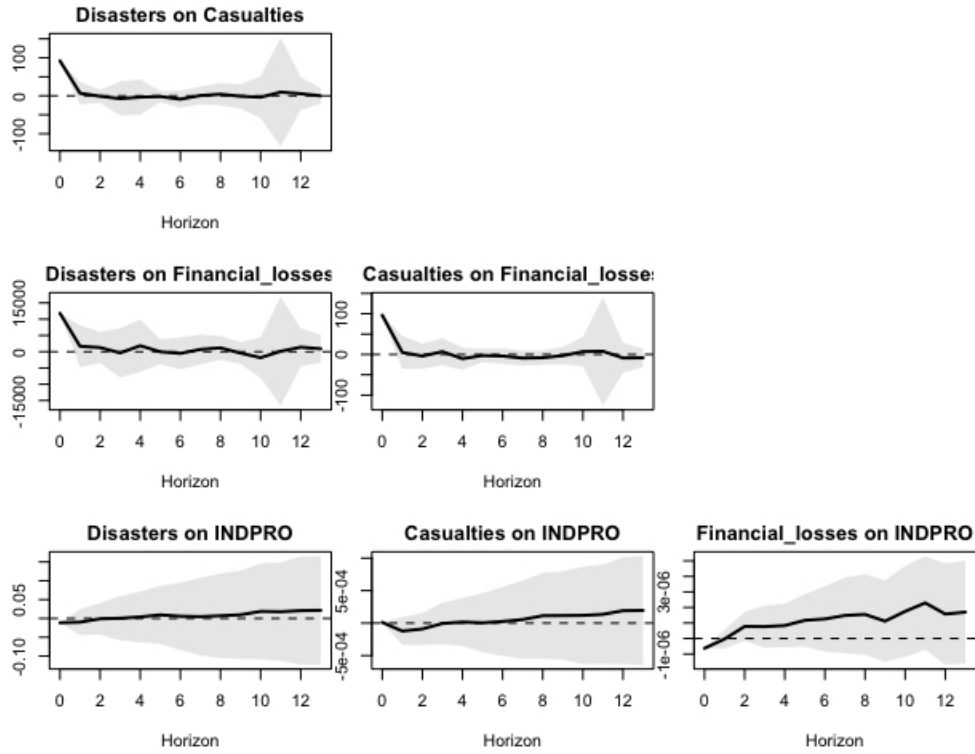


Figure 4: Impulse Responses given by a Linear Local projections

The impulse responses to disasters are constantly around zero, suggesting that the number of disasters themselves as a shock does not provoke a response in industrial production. However, casualties and financial losses have something of a more significant effect. We see an immediate negative effect of casualties, which might be explained by the properties of natural disasters, which are, in many cases, unpredictable and strike unprepared people, producing a high number of early casualties. On the other hand, financial losses impact occur more distant in the future could be the product of renewed investments with the goal of reconstruction of infrastructure. It could also be assumed that this might be a delayed effect of disasters. For example, additional financing for food industry production may occur a couple of months later. Because of the possible emptying of resources, and no possibility of resupply, due to the weather disasters. Nonetheless, the effect also depends on the length of the disaster and the way it may affect economic capital. In the end, we look into coefficients and we spot that the coefficients of the variables are somewhat significant, but the significance diminishes with the lags. There is a connection between estimate precision and confidence interval. So we could say that the increase of the width of the confidence interval suggests that there is a decrease in precision and significance. Furthermore, confidence bands suggest that the variability of predictions varies significantly over the horizons.

We can conclude that the linear model shows that disasters do not depress industrial production long-term. However, depending on the magnitude of the disaster, as we see through financial losses and casualties, there is a greater response.

6.3 Non-Linear Local Projection

The previous example only estimates impulse response functions for the linear case and diminishes the possible nonlinear effect of disasters. This is an important assumption, as it could otherwise lead our results to be misspecified. The relationship as well as the effect might not follow a linear pattern, making it uninterpretable. We are aware of non non-linear patterns in our data, from the previously conducted time series analysis.

We apply a new function `lp_np`, and we are also going to introduce the previously explained switching variable into our model. This will, in turn, help us estimate the effect of disaster in different economic environments and should help treat the outliers. We are going to present two sets of results for two different switching variables. One is the threshold based on the unemployment rate, and the other switching variable makes use of the HP-filter and logistic transition function. In the latter case, we are going to choose our gamma parameter to be 15. Following this, we again look at the results of the minimalization of BIC, and set the lag to 1 for both linear and nonlinear cases of endogenous data, as the function takes in both. This model will look for a similar immediate effect of disasters compared to its linear counterpart.

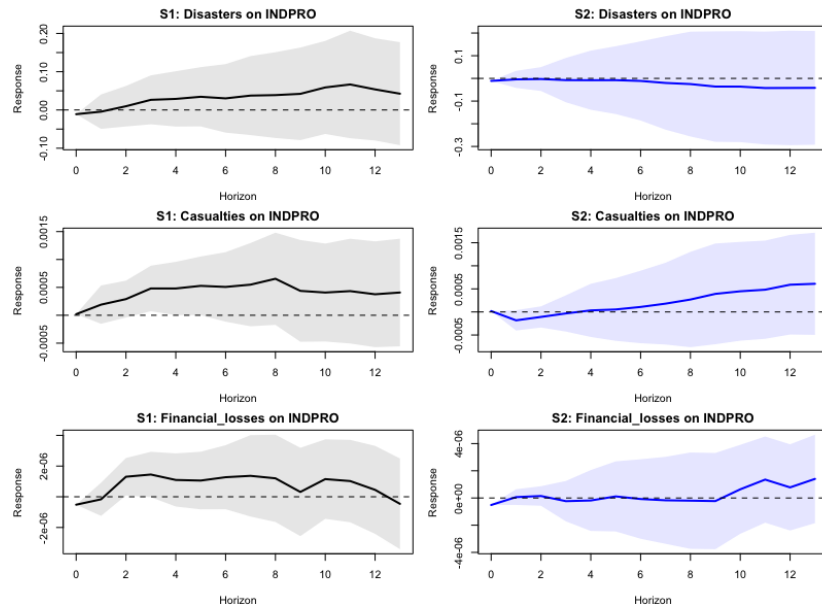


Figure 5: Impulse Responses given by a Non-Linear Local projections, and using logistic transition function

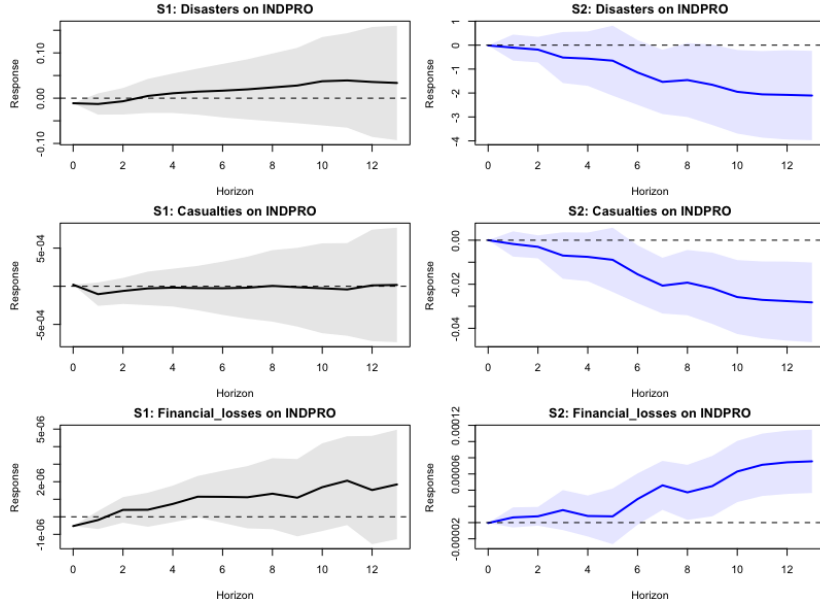


Figure 6: Impulse Responses given by a Non-Linear Local projections, and using the threshold method

Each of the figures has both black and blue impulse response functions. The black one denotes the impulse responses during expansion, while the blue one denotes the impulse responses during recession. For the case when we used the HP filter. We see that during expansion disaster shock produces a positive response function on industrial production, as we discussed in the literature chapter; this might show the rebuilding government mindset following a disaster. We again spot a wide range of confidence intervals, which only increase with the horizon. Which again suggests a potential inaccuracy in estimates. While in different regimes we see less of a response to disaster shocks, and their magnitude, is presented by financial losses and casualties. In the threshold-based approach, during the expansion, we see impulse responses to the financial cost shock have a positive trend. When the regime changes, we see a significant difference in the response to disasters and casualties. Both have negative trends and might suggest a negative impact of weather disasters on the economy. Through the human capital, as discussed in the first chapter. Furthermore, we see that impulse responses to a financial shock in a recession are positive, growing even faster with horizons compared to the expansion. It is important to note that our estimates are sensitive to regime change as well as the use of a switching variable. We see that the 95% confidence interval is the smallest in the recession regime in the threshold-based approach, hence, these estimates should be considered as the most precise ones.

7 Conclusion

This thesis contributes to the growing body of empirical research in assessing the macroeconomic impact of climate-induced weather shocks. By applying local projections to a dataset of weather disasters and industrial production in the United States.

We conclude that there exists a moderate effect of disasters on the economic performance. Particularly through casualties and financial losses, which express the magnitude of weather disasters. More so than the aggregate count of disasters itself.

The findings support the view that the disaster effect is not merely trivial. They can have delayed and regime-dependent effects, as seen by the non-linear model responses under different economic conditions. In particular, industrial production is affected more during recessions, which also further highlights its asymmetric vulnerabilities in both recession and expansion.

Nevertheless, this study is not without limitations. The use of aggregated national data masks regional disaster impact. Additionally, data conservatism and possible under-reporting, particularly when the disasters have more of a social than an economic impact. This may lead to a downward bias. Furthermore more comprehensive study considering more dependent variables could show the effect better, throughout different sectors and industries.

Further research could explore the effect of disasters on different economic sectors. As well as the disaggregated analysis of the effect of weather disasters on either state or regional level. Furthermore, possible changes of the model to include additional dependent variables, and possible spillover effects, could produce different deeper insight results.

As a statistician, it falls outside the scope of the study to talk about any policy implementation these results could suggest, from our limited dataset, and with limited knowledge of phenomena such as weather disasters. However, one could use it as a part of the bigger research to emphasize proactive economic stabilization strategies following disasters. Overall, this study argues that the effect of extreme weather disasters is not only an environmental problem but also an economic one. This paper could be used as an example, in continued research in this topic as we try to adapt to an increasingly volatile world.

8 References

Auerbach, A. J. and Gorodnichenko, Y. (2011). Fiscal Multipliers in Recession and Expansion. *National Bureau of Economic Research*, No. 17477 Retrieved from <https://doi.org/10.3386/w17447>

Acevedo, S., Mrkaic, M., Novta, N., Pugacheva, E., & Topalova, P. (2020). The effects of weather shocks on economic activity: What are the channels of impact? *Journal of Macroeconomics*, 65, 103207. Retrieved from <https://doi.org/10.1016/j.jmacro.2020.103207>

“Direct Stimulus Payments to Individuals” (2019). In *Recession Ready: Fiscal Policies to Stabilize the American Economy*, edited by Heather Boushey, Ryan Nunn, and Jay Shambaugh. Washington, DC: The Hamilton Project and the Washington Center on Equitable Growth.

Bacchiocchi, E., Bastianin, A., & Moramarco, B. (2024). Macroeconomic spillover of weather shocks across the U.S. Retrieved from <https://arxiv.org/abs/2403.10907>

Cavallo, E., Galiani, S., Noy, I., & Pantano, J. (2013). CATASTROPHIC NATURAL DISASTERS AND ECONOMIC GROWTH. *The Review of Economics and Statistics*, 95(5), 1549–1561. <http://www.jstor.org/stable/43554847>

Eickmeier, S., Quast, J., & Schöler, Y. (2024). Macroeconomic and financial effects of natural disasters. *CAMA Working Paper No. 23/2024*. Centre for Applied Macroeconomic Analysis. Retrieved from https://crawford.anu.edu.au/sites/default/files/publication/cama_crawford_anu_edu_au/2024-03/23_2024_eickmeier_quast_schuler.pdf

Ferreira, S. (2024). Extreme weather events and climate change: Economic impacts and adaptation policies. *Annual Review of Resource Economics*, 16, 207–231. Retrieved from <https://doi.org/10.1146/annurev-resource-101623-095314>

Felbermayr, G., Gröschl, J., Sanders, M., Schippers, V., Steinwachs, T. (2022). The economic impact of weather anomalies. *World Development*, 151, 105745. Retrieved from <https://doi.org/10.1016/j.worlddev.2021.105745>.

Hodrick, Robert; Prescott, Edward C. (1997). "Postwar U.S. Business Cycles: An Empirical Investigation" (PDF). *Journal of Money, Credit, and Banking*, 29(1): 1–16. doi:10.2307/2953682. JSTOR 2953682. S2CID 154995815.

Jahn, M. (2015). Economics of extreme weather events: Terminology and regional impact models. *Weather and Climate Extremes*, 10(B), 29–39. Retrieved from <https://doi.org/10.1016/j.wace.2015.08.005>

Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1), 161–182. Jordà, Ò. (2023). Local projections for applied economics. *Federal Reserve Bank of San Francisco Working Paper No. 2023-16*. <https://doi.org/10.24148/wp2023-16>

Jordà, Ò. (2023). Local projections for applied economics. *Federal Reserve Bank of San Francisco Working Paper No. 2023-16*. <https://doi.org/10.24148/wp2023-16>

Li, D., Plagborg-Møller, M., & Wolf, C. K. (2024). Local projections vs. VARs: Lessons from thousands of DGPs. *Journal of Econometrics*, 244(2), 105722. <https://doi.org/10.1016/j.jeconom.2024.105722>

Lusompa, A. (2021). Local projections, autocorrelation, and efficiency. *Federal Reserve Bank of Kansas City Working Paper No. 21-01*. <https://doi.org/10.18651/RWP2021-01>

Lorek, S. & Spangenberg, J. (2014). Sustainable consumption within a sustainable economy – beyond green growth and green economies. *Journal of Cleaner Production* 63, 33–44. Retrieved from <https://doi.org/10.1016/j.jclepro.2013.08.045>

Newbold, S.C., Marten, A. (2014). The value of information for integrated assessment models of climate change, *Journal of Environmental Economics and Management*, 68(1), Pages 111–123, ISSN 0095-0696, <https://doi.org/10.1016/j.jeem.2014.01.002>